

IVOLT

Privacy-First Technology & Security



Gmail IMAP Archiver

Operator, Configuration & Processing Guide

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1. System Overview

The IVolt Gmail IMAP Archiver is a console-driven Windows utility that connects to a Gmail account, retrieves every message with full metadata and attachments, and writes a durable, searchable, offline archive. It is built for long-running bulk capture: launch `Email_Archiver.exe`, verify a configuration, then process the mailbox under controlled throttling until it is fully archived — with safe resume if interrupted.

Transport is handled by MailKit; all message parsing and the cryptographic integrity hash are performed by IVolt's own parser, preserving the exact wire octets of every message. Full-text search is powered by an embedded Lucene.NET index.

Design principle

The archiver certifies **provenance and integrity of capture** — a SHA-256 over each message's raw octets — never the truth of the content.

2. What's New in This Release

This consolidated release documents the complete processing engine now that it is built. The following subsystems are new or newly specified in concrete form:

- **MailKit transport + custom parser split** — MailKit fetches raw octets and native Gmail extensions; the IVolt parser owns all decoding and the provenance hash.
- **Interactive engine** — startup screen, verified-config main menu, and guided connection diagnostics.
- **Scope selection** — full extract, resume, date range, new-since-last-run (UID high-water), or a specific label.
- **Persistent manifest** — keyed on X-GM-MSGID for O(1) resume, de-duplication, and instant summaries; rebuildable from the archive tree.
- **Lucene.NET search** — field-scoped over senders, attachment names, attachment content, email text, and headers.
- **Attachment text extraction** — text, PDF, Word, Excel; images indexed by name and size only.
- **Plugin framework** — reflection-loaded `I_Run_Code_Against_Configuration` implementations.

3. Architecture & Components

Responsibilities are cleanly separated: storage types are pure data; processing is stateless logic; transport is isolated behind MailKit; search and extraction are independent subsystems.

Component	Type	Responsibility
Entry point	Program / <code>Email_Archiver_Engine</code>	Startup screen, main menu, all interactive flows.
Config manager	<code>Gmail_IMAP_Management_Classes</code>	Prompt for email, locate config, DPAPI-unprotect, load, or create.

Component	Type	Responsibility
Config container	Configuration_Definition_Container	Typed settings storage; JSON round-trip.
Message container	Gmail_IMAP_Message_Container	Pure storage: raw bytes, headers, addresses, MIME parts, metadata.
Message processor	Gmail_IMAP_Message_Processors	Static parse pipeline + SHA-256 provenance hash.
Fetch engine	Gmail_Fetch_Engine	MailKit connect + fetch loop, scope, throttle, retry, checkpoint.
Archive store	Archive_Store	BYMONTH records, attachment ZIPs, manifest, IArchiveContext.
Index	Archive_Index	Lucene.NET writer/searcher, field-scoped.
Extractor	Attachment_Text_Extractor	Text/PDF/Word/Excel → indexable text.
Plugin loader	Plugin_Loader	Reflection discovery of archive plugins.

4. Data Flow (End to End)

```

Email_Archiver.exe
  |
  v
Startup: [1] Continue (open)    [2] New configuration
  |
  v
Gmail_IMAP_Management_Class: locate {sanitized}.json
  -> DPAPI Unprotect (CurrentUser) -> deserialize
  |
  v
Config verified -> MAIN MENU
  |
  v (Continue Processing)
Scope select -> MailKit connect (imap.gmail.com:993 SSL)
  |
For each candidate UID (dedup via manifest):
  MailKit FETCH summary (FLAGS, INTERNALDATE, SIZE, X-GM-*)
  MailKit GetStream BODY.PEEK[] -> raw octets
    -> Gmail_IMAP_Message_Processors.Process() (parse + SHA-256)
    -> ApplyImapMetadata() (UID, flags, labels, gm ids)
    -> Archive_Store.StoreMessage():
      BYMONTH JSON record + attachment ZIP
      extract text -> Lucene index
      update manifest (seen, counts, dates, pointer)
    -> throttle / retry / checkpoint

```

5. Prerequisites

Requirement	Detail
Operating system	Windows. Config protection uses DPAPI (ProtectedData).
.NET	net8.0-windows; Newtonsoft.Json (System.Text.Json is not used).
Libraries	MailKit, Lucene.NET (+ Analysis.Common, QueryParser), PdfPig, DocumentFormat.OpenXml.
Gmail account	IMAP enabled, 2-Step Verification on, 16-char App Password issued.
Windows account	Run under the SAME user that created the config (DPAPI CurrentUser).
Disk	Space for raw archives + attachment ZIPs + Lucene index.

6. Gmail Account Preparation

Gmail requires a Google **App Password** for IMAP when 2-Step Verification is enabled. This 16-character value is used for the IMAP, POP and SMTP secret fields.

- 1 Enable 2-Step Verification on the Google account.
- 2 Google Account → Security → App passwords → generate a new password.
- 3 Copy the 16-character value (spaces are ignored).
- 4 Provide it when the archiver prompts — it is stored only inside the DPAPI-protected config.

Security note

The App Password grants mailbox access. It is never written to disk in plaintext by this system. Revoke it in the Google console if a workstation is lost.

7. Configuration Files

One configuration file per account, located at:

```
Resources\Configurations\{email_sanitized}.json
```

The sanitized name is the address lower-cased with every @ and . replaced by an underscore — **markalicz@gmail.com** becomes **markalicz_gmail_com.json**. The file is DPAPI-protected (CurrentUser) and Base64-encoded; on load the manager reverses that and deserializes into the typed container.

8. Launching the Archiver — Startup Screen

Running Email_Archiver.exe presents the first choice:

```

=====
IVolt Gmail IMAP Archiver
Privacy-First Technology & Security · ivolt.io
=====
[1] Continue – open an existing configuration
[2] New configuration
[0] Exit
Choose: _

```

- **Continue** prompts for the email, locates the matching config, DPAPI-unprotects, and loads it.
- **New configuration** runs the interactive wizard, then protects and saves the result.

If decryption fails

A `CryptographicException` means the file was protected by a different Windows account or machine. Re-create the configuration under the account that will run the archiver.

9. The Main Menu

After a configuration is verified, the main menu appears:

```

===== MAIN MENU =====
1. Test Connections
2. Show Log / Data Summary
3. Continue Processing Configuration File
4. Search the Archive
5. Run Plugin
6. Switch Configuration
0. Exit

```

Item	Action
1. Test Connections	Verify IMAP and SMTP with guided, type-specific failure diagnostics.
2. Show Log / Data Summary	Message count, attachment count, date range, index docs — from the manifest.
3. Continue Processing	Open the scope selector and run the archival fetch loop.
4. Search the Archive	Field-scoped Lucene search over the local archive.
5. Run Plugin	Discover and run <code>I_Run_Code_Against_Configuration</code> plugins.
6. Switch Configuration	Load and verify a different account, then re-enter this menu.

Menu structure note

Search and Run Plugin are top-level operations on the active archive, so they never require re-loading a configuration first. Switching accounts is its own explicit action.

10. Test Connections (Guided Diagnostics)

Test Connections checks IMAP and SMTP independently and diagnoses failures by type, so the operator is told exactly what is wrong and how to fix it rather than seeing a raw error.

Failure	What it means	Guided fix shown
Authentication	App password wrong/revoked, or 2FA off	Generate a fresh 16-char App Password; re-run New configuration.
TLS handshake	Certificate / interception issue	Check system clock; check for a TLS-intercepting proxy or AV.
Socket / host	Server unreachable	Verify network access to the host:port; check firewall.
Success	IMAP reachable	Reports the message count in [Gmail]/All Mail.

11. Continue Processing — Scope Selection

Choosing Continue Processing opens the scope selector:

Scope	Behavior	When to use
Full extract	Every message in All Mail	First-ever capture, or a full re-verify.
Resume	Skip already-archived (manifest dedup)	Default; safe to run anytime.
Date range	Messages within Since/Before	Targeted backfill or a specific period.
New since last run	UID above the stored high-water mark	Fast incremental after a full capture.
Specific label	One Gmail label / folder	Archive a single label in isolation.

Gmail label model

Labels are exposed as IMAP folders and one message can carry several. Archiving [Gmail]/All Mail captures each message once; its labels are preserved in X-GM-LABELS rather than duplicated per folder.

12. Storage Layout, Manifest & Archive Pointer

{BASEPATH}\	
ARCHIVES\{email_sanitized}\	
2026\01\ 2026\02\ ...	(BYMONTH message records, JSON)
{sanitized}.manifest.json	(resume / dedup / summary state)
ATTACHMENTS\ -> {pointer}.zip	(per-message attachment ZIPs)
INDEX\{email_sanitized}\INDEX\	(Lucene.NET index)

The manifest

A per-account manifest is written incrementally during archival. It records the X-GM-MSGID → archive-pointer map (the dedup/skip set), running counts, the message date range, and the UID high-water mark. It gives O(1) resume and instant summaries, and is written atomically so a crash never corrupts it. If lost, it is rebuilt by scanning the archive tree.

The archive pointer

Each stored message gets a monotonically increasing archive pointer, starting from `archive_pointer_starting_increment` (default 1212). It is the durable local key that links a record to its attachment ZIP and index entry, and is independent of Gmail's UID (which can shift on UIDVALIDITY changes).

13. Indexing (Lucene.NET)

An embedded Lucene.NET index is written under `INDEX\{sanitized}\`. Each message becomes one document, added incrementally during archival and keyed on X-GM-MSGID so re-runs are idempotent. Indexed fields honor the `index_email_*` switches; a disabled field stays in the stored record but is omitted from the index.

Index field	Source	Config switch
from	Sender address/name	<code>index_email_from</code>
to / cc / bcc	Recipient lists	<code>index_email_to / _cc / _bcc</code>
subject	Decoded subject	<code>index_email_subject</code>
body	Plain-text body	<code>index_email_body</code>
headers	Flattened header set	(always)
att_names	Attachment filenames	<code>index_email_attachments</code>
att_content	Extracted attachment text	<code>index_email_attachments</code>
date	yyyyMMdd for range queries	<code>index_email_date</code>

14. Attachment Handling, Extraction & Zipping

When `archive_attachments` is enabled, each attachment part is decoded back to its original bytes and written into a per-message ZIP named by the archive pointer. During zipping, indexable text is extracted where the format allows.

Format	Extraction	Indexed
Plain text (txt, csv, log, xml, json, source, etc.)	Direct decode (BOM-aware)	Full content
PDF	PdfPig text extraction	Full text
Word (.docx)	Open XML paragraph text	Full text
Excel (.xlsx)	Open XML cells + shared strings	Cell text

Format	Extraction	Indexed
Images (png, jpg, gif, ...)	None	Name, type, size only
Other binaries	None	Name, type, size only

Honest indexing

Formats that cannot be turned into text are recorded with filename, type and size and marked `content_indexed = false`. The index never claims to have searched bytes it could not read. OCR of scanned/image PDFs is out of scope in this release.

15. Searching the Archive

Search runs entirely against the local archive — nothing is re-fetched from Gmail. Each option maps to one Lucene field; results are returned most-relevant first and shown as pointer, date, sender and subject.

```
===== SEARCH ARCHIVE =====
1. Search Senders           -> from
2. Search Attachment Names  -> att_names
3. Search Attachment Content-> att_content
4. Search Email Text        -> body
5. Search Header Data       -> headers
0. Back
```

Queries accept Lucene syntax (terms, phrases in quotes, wildcards, boolean AND/OR/NOT). Attachment-content search reaches only formats that were text-extractable at archival time.

16. Plugins

Custom logic runs through the `I_Run_Code_Against_Configuration` interface. The loader discovers every active implementation in loaded assemblies and in a `Plugins` folder beside the executable, orders them by `Execution_Order`, and offers them under `Run Plugin`. Each plugin receives the loaded configuration, the archive context (records, manifest, attachments, index), and a console writer for progress.

```
public interface I_Run_Code_Against_Configuration {
    string Plugin_Name { get; }
    string Plugin_Description { get; }
    int Execution_Order { get; }
    bool Active { get; }
    bool Run(Configuration_Definition_Container config,
              IArchiveContext archive, TextWriter output);
}
```

17. Throttling, Retries & Resumability

Setting	Default	Effect
max_emails_per_minute	100	Fetch rate ceiling per rolling minute.
max_attachment_size_in_MB_per_minute	50	Download-volume ceiling per rolling minute.
pause_between_requests_in_seconds	11	Fixed delay between messages.
max_retries	3	Attempts per message with backoff before skipping.
log_level	DETAIL	Operation log verbosity.

The loop checkpoints the index and manifest roughly every 25 messages, so an interruption loses at most a handful of messages of progress. Because dedup is keyed on X-GM-MSGID, resuming simply skips what is already archived.

18. Security Model

- **At rest:** the App Password lives only inside the DPAPI-protected config, bound to the Windows user.
- **In transit:** IMAP/SMTP use SSL/TLS.
- **At the prompt:** the password is masked so it never appears in console scrollback.
- **Integrity:** SHA-256 over raw octets makes post-hoc tampering detectable.
- **Provenance-preserving fetch:** BODY.PEEK[] retrieves raw bytes without marking messages read.
- **Least exposure:** sensitive index fields (e.g. bcc) can be disabled without losing the stored record.

19. Operating Procedures

Initial full archive

- 1 Prepare the Gmail App Password (Section 6).
- 2 Launch; choose New configuration; complete the wizard.
- 3 Run Test Connections and confirm IMAP + SMTP pass.
- 4 Choose Continue Processing → Full extract; let it run under throttling.
- 5 Spot-check ARCHIVES, ATTACHMENTS and INDEX, then Show Summary.

Recurring incremental archive

- 1 Launch; choose Continue; enter the same email (config loads automatically).
- 2 Continue Processing → New since last run (or Resume).
- 3 Confirm the summary counts and date range advanced.

20. Troubleshooting

Symptom	Likely cause	Resolution
CryptographicException on load	Config protected by another user/machine	Recreate the config under the running account.
IMAP/SMTP auth rejected	App Password wrong/revoked or 2FA off	Issue a fresh App Password; re-run New configuration.
Connection dropped mid-run	Gmail rate limiting	Lower max_emails_per_minute; raise the pause; resume.
Attachment content not found	Format not text-extractable (image/binary)	Expected; only text/PDF/Word/Excel content is indexed.
Summary shows zero	Nothing archived yet, or manifest lost	Run Continue Processing; if lost, the manifest rebuilds from the tree.
Duplicate-looking hits	Same message under multiple labels	Archive All Mail; dedup is by X-GM-MSGID.

21. Glossary

Term	Definition
App Password	16-char Google credential for IMAP under 2-Step Verification.
Archive pointer	IVolt-assigned increasing integer identity for a stored message.
BODY.PEEK[]	IMAP fetch of raw octets without setting the \Seen flag.
BYMONTH	Storage grouping of records into year/month folders.
DPAPI	Windows Data Protection API; binds the config to a user/machine.
Manifest	Per-account state file: dedup map, counts, dates, UID high-water.
Provenance	Verifiable record of how data was captured — not a truth claim.
X-GM-MSGID / THRID	Gmail proprietary 64-bit message / thread identifiers.
X-GM-LABELS	Gmail proprietary list of labels on a message.

Appendix A — Source File Inventory

File	Role
Program.cs	Executable entry point (Email_Archiver.exe).
Email_Archiver_Engine.cs	Startup, main menu, diagnostics, summary, search UI, plugin UI.
Gmail_Fetch_Engine.cs	MailKit transport, fetch loop, scope, throttle, retry, checkpoint.
Archive_Store.cs	BYMONTH records, attachment ZIPs, manifest, IArchiveContext.

File	Role
Archive_Index.cs	Lucene.NET writer/searcher.
Archive_Model.cs	Manifest, record, attachment models, IArchiveContext.
Attachment_Text_Extractor.cs	Text/PDF/Word/Excel extraction.
Plugin_Loader.cs	Reflection plugin discovery.
I_Run_Code_Against_Configuration.cs	Plugin contract.
Gmail_IMAP_Message_Container.cs	Message storage (prior turn).
Gmail_IMAP_Message_Processors.cs	Message parser + SHA-256 (prior turn).
Configuration_Definition_Container.cs	Config storage (prior turn).
Gmail_IMAP_Management_Class.cs	Config load/create + DPAPI (prior turn).

Appendix B — Configuration Field Reference

Root fields

Field	Type	Purpose
email	string	Account address; IMAP login and SMTP username.
password	string	Account / app password (secret).
server	object	IMAP/POP/SMTP endpoints and secrets.
performance	object	Throttling and retry settings.
archive_index_folder	string	Lucene index destination.
archive_folder	string	Message-record destination.
archive_structure	string	Grouping scheme (BYMONTH).
archive_format	string	Stored record format (JSON).
archive_attachments	bool	Extract and store attachments.
archive_pointer_starting_increment	long	First archive pointer (default 1212).
archive_attachments_format	string	Attachment container (ZIP).
archive_attachments_folder	string	Attachment ZIP destination.
index_fileformat_type	string	Index record format (IVOLT_SCAN).
index_email_body ... _attachments	bool	Per-field index toggles (default true).

performance

Field	Default	Purpose
max_emails_per_minute	100	Fetch rate ceiling.
max_attachment_size_in_MB_per_minute	50	Attachment download ceiling.
pause_between_requests_in_seconds	11	Inter-request delay.
max_retries	3	Retries per message before skipping.
log_level	DETAIL	Log verbosity.

IVolt LLC — Privacy-First Technology & Security. Confidential; for authorized operators. The archiver certifies provenance and integrity of capture, not the truth of message content.